

Activity 01

Customers Table

```
> desc Customers
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI		
FullName	varchar(100)	YES			
PhoneNumber	varchar(15)	YES			

Waiters Table

```
> desc Waiters
```

Field	Type	Null	Key	Default	Extra
WaiterID	int	NO	PRI		
Name	varchar(100)	YES			
Shift	varchar(20)	YES			

MenuItems Table

```
> desc MenuItems
```

Field	Type	Null	Key	Default	Extra
ItemID	int	NO	PRI		
DishName	varchar(100)	YES			
Price	decimal(10,2)	YES			

Orders Table

```
> desc Orders
```

Field	Type	Null	Key	Default	Extra
OrderID	int	NO	PRI		
OrderDate	date	YES			
CustomerID	int	YES	MUL		
WaiterID	int	YES	MUL		

OrderDetails Table

```
> desc OrderDetails
```

Field	Type	Null	Key	Default	Extra
DetailID	int	NO	PRI		
OrderID	int	YES	MUL		
ItemID	int	YES	MUL		
Quantity	int	YES			

Write the SQL queries for the following questions. Every question here requires joining exactly two tables based on the schema you provided.

1. Customer Orders: Write a query to display the customer's FullName and the OrderDate for every order placed at the restaurant. (Hint: Join Customers and Orders tables).

```
SELECT Customers.FullName, Orders.OrderDate
FROM Customers
INNER JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

FullName	OrderDate
Kamal Perera	2023-10-25
Nimal Silva	2023-10-25
Sunil Fernando	2023-10-26

2. Waiter Service: Write a query to list all OrderIDs and the Name of the waiter who served each specific order. (Hint: Join Orders and Waiters tables).

```
SELECT Orders.OrderID, Waiters.Name
FROM Orders
INNER JOIN Waiters
ON Orders.WaiterID = Waiters.WaiterID;
```

OrderID	Name
101	Saman
103	Saman
102	Ruwan

3. Dish Quantities: Write a query to show the DishName from the menu and the Quantity ordered for each item in the order details. (Hint: Join MenuItems and OrderDetails tables).

```
SELECT MenuItems.DishName, OrderDetails.Quantity
FROM MenuItems
INNER JOIN OrderDetails
ON MenuItems.ItemID = OrderDetails.ItemID;
```

DishName	Quantity
Rice and Curry	2
Fried Rice	1
Kottu	1
Burger	2
Soft Drink	2
Soft Drink	3

4. Morning Shift Orders: Write a query to find the OrderIDs handled by waiters working the "Morning" shift, along with the waiter's Name. (Hint: Join Waiters and Orders tables, and use a WHERE condition for the shift).

```
SELECT Orders.OrderID, Waiters.Name
FROM Orders
INNER JOIN Waiters
ON Orders.WaiterID = Waiters.WaiterID
WHERE Waiters.Shift = 'Morning';
```

OrderID	Name
101	Saman
103	Saman

5. Customer Search: Write a query to display the FullName and PhoneNumber of customers, along with their OrderID, but only for orders placed on '2023-10-25' (or any specific date you used in your dummy data). (Hint: Join Customers and Orders tables).

```
SELECT Customers.FullName,
       Customers.PhoneNumber,
       Orders.OrderID
FROM Customers
INNER JOIN Orders
ON Customers.CustomerID = Orders.CustomerID
WHERE Orders.OrderDate = '2023-10-25';
```

FullName	PhoneNumber	OrderID
Kamal Perera	0711111111	101
Nimal Silva	0722222222	102

Activity 02

Students table

```
> desc Students
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI		
StudentName	varchar(100)	YES			
Department	varchar(100)	YES			

Categories table

```
> desc Categories
```

Field	Type	Null	Key	Default	Extra
CategoryID	int	NO	PRI		
CategoryName	varchar(100)	YES			

Books table

```
> desc Books
```

Field	Type	Null	Key	Default	Extra
BookID	int	NO	PRI		
BookTitle	varchar(100)	YES			
CategoryID	int	YES	MUL		

BorrowRecords table

```
> desc BorrowRecords
```

Field	Type	Null	Key	Default	Extra
RecordID	int	NO	PRI		
StudentID	int	YES	MUL		
BookID	int	YES	MUL		
BorrowDate	date	YES			

Based on the scenario above, write the SQL queries for the following questions

1. Write a query to display all BookTitles along with their corresponding CategoryName.

```
SELECT Books.BookTitle, Categories.CategoryName
FROM Books
INNER JOIN Categories
ON Books.CategoryID = Categories.CategoryID;
```

BookTitle	CategoryName
Java Programming	Programming
Database Systems	Programming
Physics Fundamentals	Science
World History	History

2. Write a query to list the StudentName and the BorrowDate for every time a student has borrowed a book.

```
SELECT Students.StudentName, BorrowRecords.BorrowDate
FROM Students
INNER JOIN BorrowRecords
ON Students.StudentID = BorrowRecords.StudentID;
```

StudentName	BorrowDate
Nimal Perera	2024-01-10
Nimal Perera	2024-01-15
Kamal Silva	2024-01-12
Saman Fernando	2024-01-18

3. Write a query to display the StudentName and the BookTitle of the books they have borrowed. (Hint: You need to link the student to the record, and the record to the book).

```

SELECT Students.StudentName, Books.BookTitle
FROM BorrowRecords
INNER JOIN Students
ON BorrowRecords.StudentID = Students.StudentID
INNER JOIN Books
ON BorrowRecords.BookID = Books.BookID;

```

StudentName	BookTitle
Nimal Perera	Java Programming
Nimal Perera	Database Systems
Kamal Silva	Physics Fundamentals
Saman Fernando	World History

4. Write a query to find the BookTitle, BorrowDate, and CategoryName for all borrowed books.

```

SELECT Books.BookTitle,
       BorrowRecords.BorrowDate,
       Categories.CategoryName
FROM BorrowRecords
INNER JOIN Books
ON BorrowRecords.BookID = Books.BookID
INNER JOIN Categories
ON Books.CategoryID = Categories.CategoryID;

```

BookTitle	BorrowDate	CategoryName
Java Programming	2024-01-10	Programming
Database Systems	2024-01-15	Programming
Physics Fundamentals	2024-01-12	Science
World History	2024-01-18	History

Activity 03

```
> desc Orders
```

Field	Type	Null	Key	Default	Extra
OrderID	int	NO	PRI		
CustomerID	int	YES	MUL		
OrderDate	date	YES			
TotalAmount	decimal(10,2)	YES			

```
> desc Products
```

Field	Type	Null	Key	Default	Extra
ProductID	int	NO	PRI		
ProductName	varchar(100)	YES			
Price	decimal(10,2)	YES			
CategoryID	int	YES	MUL		

```
> desc Categories
```

Field	Type	Null	Key	Default	Extra
CategoryID	int	NO	PRI		
CategoryName	varchar(100)	YES			


```
> desc Customers
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI		
CustomerName	varchar(100)	YES			
City	varchar(50)	YES			

Write the SQL queries for the following questions Every question here requires joining exactly two tables.

1.Customer Orders: Write a query to display the CustomerName, the OrderID, and the OrderDate for every order placed. (Hint: Join Customers and Orders).

```
SELECT Customers.CustomerName,
```

```
       Orders.OrderID,
```

```
       Orders.OrderDate
```

```
FROM Orders
```

```
INNER JOIN Customers
```

```
ON Orders.CustomerID = Customers.CustomerID;
```

CustomerName	OrderID	OrderDate
Kamal Perera	1001	2024-02-10
Nimal Silva	1002	2024-02-11
Sunil Fernando	1003	2024-02-12

2.Product Categories: Write a query to list all ProductNames, their Price, and their corresponding CategoryName. (Hint: Join Products and Categories).

```
SELECT Products.ProductName,  
  
       Products.Price,  
  
       Categories.CategoryName  
  
FROM Products  
  
INNER JOIN Categories  
  
ON Products.CategoryID = Categories.CategoryID;
```

ProductName	Price	CategoryName
Laptop	150000.00	Electronics
Smartphone	80000.00	Electronics
T-Shirt	2500.00	Clothing
Database Book	3500.00	Books

3.High-Value Orders: Write a query to show the CustomerName and the TotalAmount of their order, but only for orders where the TotalAmount is greater than 1000.

```
SELECT Customers.CustomerName,  
  
       Orders.TotalAmount  
  
FROM Orders  
  
INNER JOIN Customers  
  
ON Orders.CustomerID = Customers.CustomerID  
  
WHERE Orders.TotalAmount > 1000;
```

CustomerName	TotalAmount
Kamal Perera	150000.00
Nimal Silva	2500.00
Sunil Fernando	3500.00

4.City Filter: Write a query to find the CustomerName and the OrderID for customers who live in the city of 'Colombo'.

```

SELECT Customers.CustomerName,
       Orders.OrderID
FROM Orders
INNER JOIN Customers
ON Orders.CustomerID = Customers.CustomerID
WHERE Customers.City = 'Colombo';

```

CustomerName	OrderID
Kamal Perera	1001
Sunil Fernando	1003

Activity 04

```
> desc Employees
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI		
EmpName	varchar(100)	YES			
DeptID	int	YES	MUL		

```
> desc Departments
```

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI		
DeptName	varchar(100)	YES			

```
> desc CompanyCars
```

Field	Type	Null	Key	Default	Extra
CarID	int	NO	PRI		
CarModel	varchar(100)	YES			
EmpID	int	YES	MUL		

Write the SQL queries for the following questions. Read the instructions carefully to know which type of join to use.

1.Standard Employee List: Write a query to display the EmpName and their corresponding DeptName. *Condition:* **Only** include employees who are already assigned to a department.

Hint: Use an INNER JOIN between Employees and Departments.

```
SELECT Employees.EmpName,  
       Departments.DeptName  
FROM Employees  
INNER JOIN Departments  
ON Employees.DeptID = Departments.DeptID;
```

EmpName	DeptName
Kamal Perera	IT
Nimal Silva	HR
Sunil Fernando	Finance

2. Car Assignments: Write a query to list **all** EmpNames and the CarModel of the company car assigned to them.

- *Condition:* If an employee does not have a company car, their name must **still appear** in the list (with the car model showing as NULL).
- *Hint:* Use a LEFT JOIN starting with the Employees table on the left, joining with the CompanyCars table.

```
SELECT Employees.EmpName,
       CompanyCars.CarModel
FROM Employees
LEFT JOIN CompanyCars
ON Employees.EmpID = CompanyCars.EmpID;
```

EmpName	CarModel
Kamal Perera	Toyota Corolla
Nimal Silva	
Sunil Fernando	Honda Civic
Saman Jayasuriya	

3.All Employees & Departments: Write a query to show **all** EmpNames and their DeptName. *Condition:* Even if a new employee is not yet assigned to a department, they must still be shown in the results.

Hint: Use a LEFT JOIN starting with the Employees table.

```
SELECT Employees.EmpName,
       Departments.DeptName
FROM Employees
LEFT JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

EmpName	DeptName
Kamal Perera	IT
Nimal Silva	HR
Sunil Fernando	Finance
Saman Jayasuriya	

4. **Department Check:** Write a query to display the EmpName and the DeptName.

- *Condition:* You need to make sure **every** department is listed in the results, even if there are absolutely zero employees working in that department right now.
- *Hint:* Write the query starting with Employees as your first table, and use a RIGHT JOIN to connect it to the Departments table.

```
SELECT Employees.EmpName,
       Departments.DeptName
FROM Employees
RIGHT JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

EmpName	DeptName
Kamal Perera	IT
Nimal Silva	HR
Sunil Fernando	Finance
	Marketing

5. Write a query to display the CarModel and the EmpName of the person driving it. *Condition:* You want to see **all** company cars in the database, even if a car is currently parked and not assigned to any employee.

Hint: Start your query with the Employees table, and use a RIGHT JOIN to the CompanyCars table.

```
SELECT CompanyCars.CarModel,  
       Employees.EmpName  
FROM Employees  
RIGHT JOIN CompanyCars  
ON Employees.EmpID = CompanyCars.EmpID;
```

CarModel	EmpName
Toyota Corolla	Kamal Perera
Honda Civic	Sunil Fernando
Nissan Sunny	